Performance of regression and FFN models to detect drug-protein interactions in simulated chemoproteomics data

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Abstract

Chemoproteomics enables large-scale analysis of drug-protein interactions, facilitating drug discovery and target validation. The lack of a standardized statistical analysis approach complicates the accurate identification of drug-protein interactions. This project aims to benchmark various methods used in chemoproteomics data analysis by using simulated datasets with varying levels of experimental complexity. By comparing four-parameter logistic regression, isotonic regression, and feedfoward neural networks, this project seeks to determine the most robust method. The findings will provide valuable insights into best practices for chemoproteomics data analysis.

1 Introduction

Chemoproteomics is an emerging field that measures the interactions between small molecules and the proteome, playing a crucial role in drug discovery [1]. This approach allows researchers to analyze numerous compounds and proteins simultaneously, aiding in target validation, novel target identification, and off-target effect detection.

Currently, there is no standardized statistical approach for analyzing chemoproteomics data. One of the primary difficulties in evaluating drug-protein interactions is the absence of ground truth as researchers cannot definitively confirm whether a given protein interacts with a drug. Various computational methods have been employed to infer these interactions, but their performance is difficult to assess without known ground truth data[5, 6, 4] .

To address this issue, this project will use simulated datasets to evaluate the robustness of different methods for drug-protein interaction detection. By systematically assessing the performance of existing analytical approaches under controlled conditions, I will aim to identify the most reliable methods for real-world data analysis. The goal is to compare different methods, and identify their advantages and limitations, and provide recommendations for best practices. This benchmark study will serve as a foundation for future development of chemoproteomics data analysis pipelines.

The standard approach for analyzing drug response in chemoproteomics experiments involves modeling individual drug-protein relationships using a four-parameter logistic regression model. While effective under ideal conditions, these methods often fail when faced with outliers, high biological and technical variability, and non-standard experimental designs. This project seeks to evaluate the effectiveness of different experimental designs and identify the best method dose-response analysis that is more robust to these challenges.

1.1 Related Work

Several software packages have been developed for dose-response modeling, focusing on parametric curve fitting, IC_{50} estimation, and statistical testing. Among these, the R package `drc` [5] is widely used for fitting models such as the four-parameter log-logistic curve:

$$Y_i = \theta_1 + \frac{\theta_2 - \theta_1}{1 + \left(\frac{x_i}{\theta_3}\right)^{\theta_4}} + \epsilon_i$$

where θ_1 and θ_2 are the lower and upper asymptotes, θ_3 represents the IC_{50}, θ_4 the slope, and ϵ_i is a normally distributed error term. Parameters are estimated using nonlinear least squares.

The `dr4pl` R package [6] builds upon `drc` by improving numerical stability and convergence during optimization, while retaining the same model structure and parameter interpretation. Confidence intervals for IC_{50} and slope parameters are provided to assess curve reliability.

`CurveCurator` [4], implemented in Python, identifies significant curves using a recalibrated F-statistic and effect size comparison between the highest drug dose and control. It also uses the 4PL model as its basis.

Other relevant tools include `DoseFinding`, which supports model-based dose-finding with both frequentest and Bayesian approaches, and `nlstools`, which aids in the evaluation of nonlinear least squares fits through diagnostics and confidence interval calculations.

While these tools are widely used, many do not explicitly account for biological variability unless replicates are available. In single-replicate settings, uncertainty in IC_{50} estimates may be underestimated, as only lack-of-fit error is modeled. Our simulation framework enables benchmarking of these methods under realistic assumptions, including replicate structure and outlier contamination.

2 Methods

This analysis uses a simulated chemoproteomic dataset representing protein-level responses to increasing concentrations of a drug. The simulated dataset includes log-transformed intensity values for each protein across nine drug concentrations (DMSO + 8 increasing doses) and labels indicating strong, weak, or non-interacting proteins. For each protein i and dose level j, the measured response is denoted as Y_{ij}.

The goal of the analysis is to evaluate statistical and machine learning methods for detecting drug-protein interactions based on these dose-response patterns. We evaluated three approaches: a four-parameter logistic regression model[3], isotonic regression[2], and a feedforward neural network (FNN). Each method is summarized in detail below.

2.1 Input Normalization

For the four-parameter logistic model and the FNN, relative responses are normalized to the average DMSO response for each protein:

$$R_{ij} = \frac{Y_{ij}}{\bar{Y}_{i,\mathrm{DMSO}}}$$

where $\bar{Y}_{i,\mathrm{DMSO}}$ is the mean log-intensity of protein i under the DMSO (control) condition. For isotonic regression, the log-intensity values Y_{ij} were used directly.

2.2 Four-Parameter Logistic Regression

The four-parameter logistic (4PL) model assumes a sigmoidal relationship between drug dose and protein response. The model is defined as:

$$f(x) = d + \frac{a - d}{1 + \left(\frac{x}{c}\right)^b}$$

where: x is the drug dose (log-transformed), a is the minimum response (asymptote), d is the maximum response, c is the inflection point (e.g., IC_{50}), and b is the slope.

Model fitting was performed using the `lmfit` Python package. An F-test was used to compare the 4PL model to a null model (constant response) for significance testing. The test statistic is computed as:

$$F = \frac{(\mathrm{SSE}_{\mathrm{null}} - \mathrm{SSE}_{\mathrm{model}})/(p-1)}{\mathrm{SSE}_{\mathrm{model}}/(n-p)}$$

where $\mathrm{SSE}_{\mathrm{model}}$ and $\mathrm{SSE}_{\mathrm{null}}$ are the residual sums of squares for the fitted and null models, p is the number of parameters in the 4PL model, and n is the number of data points.

2.3 Isotonic Regression

Isotonic regression is a non-parametric technique that fits a monotonic (non-increasing) curve to the dose-response data without assuming a specific shape. Given sorted doses x_j and corresponding responses Y_{ij}, the fitted response values $\hat{Y}_{ij}$ are constrained such that:

$$\hat{Y}_{i1} \geq \hat{Y}_{i2} \geq \cdots \geq \hat{Y}_{i9}$$

The model was implemented using the `sklearn.isotonic` regression function. Goodness-of-fit was evaluated by comparing the sum of square errors (SSE) of the isotonic model to that of a null model using the F-statistic defined above.

2.4 Feedforward Neural Networks (FNNs)

The FNN model was trained using individual protein dose-response vectors of length 9 (corresponding to each dose). Each replicate was treated as a separate training sample, resulting in a total of $n = d \times r$ samples, where d is the number of dose points and r is the number of replicates per protein. The input features for the FNN were the relative response values R_{ij}.

The network architecture consisted of a single hidden layer with ReLU activation:

$$h = \mathrm{ReLU}(W_1 x + b_1)$$

followed by a sigmoid output layer for binary classification:

$$\hat{y} = \sigma(W_2 h + b_2)$$

Training was performed using the binary cross-entropy loss:

$$\mathcal{L}(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

The Adam algorithm was used for optimization. The dataset was split at the protein level to prevent data leakage. Performance was evaluated using ROC AUC, confusion matrices, and additional classification metrics.

3 Simulated Datasets

To evaluate statistical and machine learning methods under controlled conditions, I developed a simulation framework that mimics the structure and variability observed in real-world label-free chemoproteomic experiments. The simulation generates protein-level dose-response data across nine concentrations (DMSO + 8 increasing drug doses), with three types of interaction profiles: strong, weak, and non-interacting proteins.

Each simulation is defined by the number of proteins (N_{proteins}), proportions of strong, weak, and non-interactors (TP, TW, and TN), the number of replicates per dose (r), and the desired technical variability (σ^2_{tech}). The input also includes a real template for each interaction type, specifying dose-specific mean log-intensities that are used to generate realistic base curves.

For each protein, replicate-level intensity values are generated by sampling from a normal distribution:

$$Y_{ijk} = \mu_{ij} + \epsilon_{ijk}, \quad \epsilon_{ijk} \sim \mathcal{N}(0, \sigma^2_{\text{tech}})$$

where i indexes the protein, j indexes the dose, and k indexes the replicate. μ_{ij} is the base log-intensity from the template for that interaction type.

Outliers are introduced to simulate measurement artifacts. A proportion p_{outlier} of data points are selected at random (with at least one per strong/weak group), and their log-intensity values are perturbed by a log-normal random offset:

$$Y^*_{ijk} = Y_{ijk} + s \cdot \text{offset}, \quad \text{offset} \sim \text{LogNormal}(0, 1), \quad s \in \{-1, 1\}$$

The final simulated dataset includes columns for dose, replicate, log-intensity, protein name, protein group, and a binary outlier label. Sample groups are labeled as *DMSO* for the 0 concentration and *treated* for all other doses.

This simulation framework allows flexible evaluation of modeling techniques under varying conditions of technical variability, replicates, and outlier contamination. It also enables the generation of protein-level dose-response matrices suitable for input to curve-fitting, statistical testing, or deep learning models.

4 Results

4.1 Simulated Data Reflect Realistic Chemoproteomics Challenges

To evaluate method performance under conditions that resemble real-world experiments, six simulated datasets were created incorporating key challenges observed in chemoproteomics. Real datasets are often characterized by high technical and biological variability, outliers, poor detection for low abundant measurements, and a highly imbalanced distribution of interacting versus non-interacting proteins.

Each simulated dataset consists of 5,000 proteins, with 0.1% labeled as strong interactors, 0.9% as weak interactors, and the remaining 99% as non-interacting proteins. This reflects the typical class imbalance in chemoproteomics screens, where true interaction signals are rare and easily masked by noise.

Table 1 summarizes the characteristics of each dataset. The number of replicates, level of technical and biological variation, presence of outliers, were varied to create diverse evaluation scenarios. These conditions were chosen to probe the robustness of each method under practical limitations such as sparse sampling, increased noise, or data corruption.

Dataset	**Replicates**	**Variation ($\sigma^2_{\mathbf{tech}}$)**	**Outliers**	**Doses**
SimData1	3	0.27	0%	DMSO + 8 doses
SimData2	3	0.27	5%	DMSO + 8 doses
SimData3	3	0.80	0%	DMSO + 8 doses
SimData4	3	0.27	0%	DMSO, 1nM, 1000nM, 3000nM
SimData5	2	0.80	0%	DMSO + 8 doses
SimData6	1	0.80	0%	DMSO + 8 doses

Table 1: Summary of simulated datasets used to benchmark model robustness across replicate number, variability, dose design, and outlier presence.

4.2 Performance Comparison Across Models

Each of the three evaluated methods, Isotonic Regression (IR), Four-Parameter Logistic Regression (4PL), and the Feedforward Neural Network (FNN), was applied to the simulated datasets outlined in Table 1. Interacting proteins are defined as number of proteins with p-values 0.05), false discovery rate (FDR)-adjusted hits for IR and 4PL and predicted probability class agreement for FNN (i.e. $p > 0.5$ is predicted as an interaction). All performance metrics are included in Table 2

Isotonic Regression showed strong sensitivity to both replicate number and variability. In the low-noise, high-replicate condition (SimData1), it detected 33 significant proteins after FDR correction, including all 5 strong interactors and 28 weak ones. However, performance dropped considerably in high-variability (SimData3) or low-replicate (SimData5, SimData6) settings. IR could not be applied to SimData6 due

Dataset	Model	Accuracy	Precision	Recall	F1 Score
SimData1	Isotonic Regression (IR)	0.9960	0.917	0.660	0.767
	4-Parameter Logistic (4PL)	0.9948	0.929	0.520	0.667
	Feedforward Neural Network (FNN)	0.9953	1.000	0.533	0.696
SimData2	IR	0.9938	1.000	0.380	0.551
	4PL	0.9928	1.000	0.280	0.438
	FNN	0.9930	1.000	0.100	0.182
SimData3	IR	0.9910	1.000	0.100	0.182
	4PL	0.9914	0.667	0.080	0.143
	FNN	0.9900	0.000	0.000	0.000
SimData4	IR	0.9926	0.870	0.400	0.545
	4PL	0.9910	1.000	0.100	0.182
	FNN	–	–	–	–
SimData5	IR	0.9906	1.000	0.060	0.113
	4PL	0.9908	0.500	0.040	0.073
	FNN	0.9900	0.000	0.000	0.000
SimData6	IR	–	–	–	–
	4PL	0.9908	0.667	0.040	0.075
	FNN	0.9900	0.000	0.000	0.000

Table 2: Model performance across simulated datasets. Accuracy, precision, recall, and F1 score are calculated based on identification of strong and weak interactors as positives, and non-interactors as negatives. Dashes (–) indicate the model could not be applied due to data constraints.

to the absence of replicate structure, and detected only 3 proteins after correction in SimData5.

Four-Parameter Logistic Regression was more consistent across datasets but showed vulnerability to false positives in noisy or undersampled settings. For example, in SimData1 and SimData2, it returned a high number of nominally significant proteins (178 and 194), but only 26 and 14 passed FDR adjustment, respectively. The model retained high sensitivity to strong interactors but often failed to recover weak interactors in high-variance datasets (e.g., SimData3 and SimData5), identifying few or no weak hits post-correction.

Feedforward Neural Network (FNN) models demonstrated high specificity, particularly in datasets with sufficient sample size and balance (e.g., SimData1), where it identified all strong interactors and nearly half of the weak ones (13/27). However, the FNN's performance declined sharply under high noise and low replicate conditions (SimData3–6), and it failed entirely in SimData4 due to incompatible input shape caused by non-standard dose design. In these harder settings, hit rates dropped to zero across all interaction groups, reflecting limited robustness without sufficient training diversity.

4.3 Recall Comparison Across Models

Recall is the most important metric for this analysis, as other metrics will be inflated by the extreme class imbalance. Recall measures the proportion of true interacting proteins correctly identified by each method within a given interaction group (strong, weak, or non-interacting). In real experiments, misses a true-positive is more costly than predicting a false-positive.

Figure 1 summarizes the recall values across all datasets and protein groups. Each subplot corresponds to a simulated dataset, with bars grouped by interaction class and color-coded by model. Recall for strong interactors was consistently high for IR and 4PL, even under moderate noise (SimData1, SimData2), but declined in low-replicate or high-variance conditions (SimData5, SimData6). Weak interactors were notably harder to recover, with FNN showing relatively higher specificity in low-noise settings (SimData1), but failing to generalize in noisy or sparse datasets. IR and 4PL recovered weak interactors with varying success, depending on the replicate and noise levels.

Non-interacting proteins (false positives) were rarely predicted as hits by any model, indicating good specificity overall. However, some false positives were observed for IR and 4PL in high-variance settings (e.g., SimData4 and SimData3), likely due to overfitting or poor curve separation.

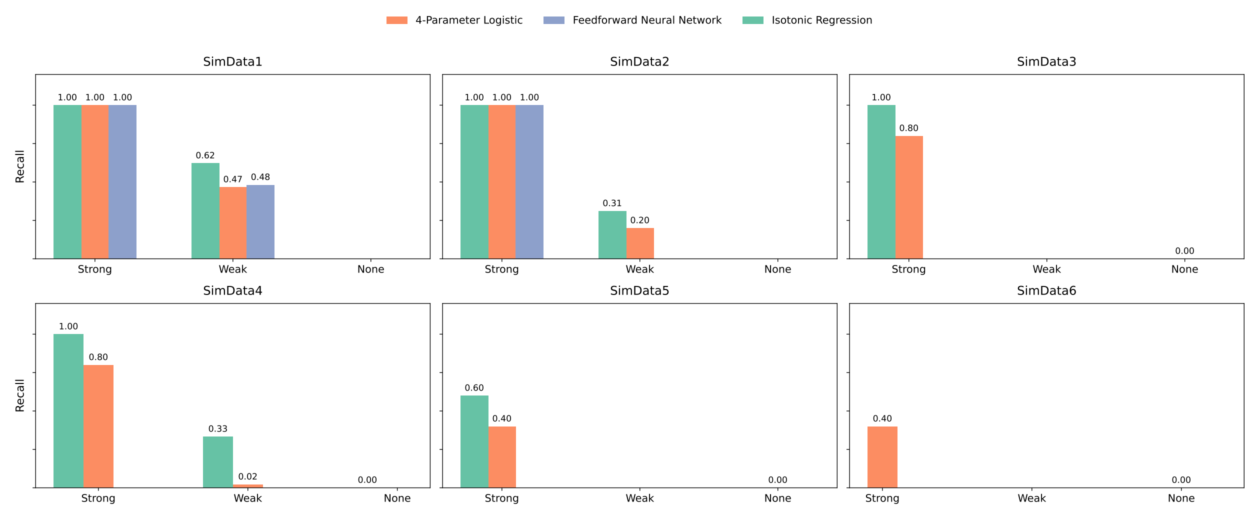

Figure 1: Recall across interaction groups (Strong, Weak, None) by model and dataset. Each panel shows one simulated dataset, with bars representing recall per protein group, color-coded by method.

5 Discussion

The results of this benchmark study highlight the complex challenges associated with analyzing chemoproteomics data, particularly in the context of drug-target interaction prediction. One of the primary difficulties is the extreme class imbalance typical of such experiments. In a realistic setting, a compound may bind strongly to only a handful of targets (e.g. 5 proteins) with an additional 10 to 40 proteins showing weaker interactions. These true positive signals are embedded within a background of thousands of non-interacting proteins, leading to highly imbalanced classification tasks where standard models may struggle to recover rare but biologically meaningful hits.

Deep learning models, such as the feedforward neural network (FNN) evaluated here, face substantial obstacles in this setting. While these models have the potential to learn complex, non-linear patterns across protein response profiles, they also require large, balanced datasets to perform optimally. In chemoproteomics, however, the number of samples is typically small compared to the number of predictors. Even with regularization and adjustments for class imbalance, the FNN model showed limited ability to generalize in noisy or low-replicate conditions. Furthermore, low-abundance proteins—which are often of biological interest—may be systematically underdetected in mass spectrometry-based experiments, further reducing model visibility into the true interaction landscape.

Traditional statistical approaches like isotonic regression (IR) and the four-parameter logistic (4PL) model treat each protein independently. This independence assumption has both benefits and drawbacks. On one hand, it allows for localized modeling that is robust to outliers in other proteins; on the other, it discards information that may be shared across the proteome, such as global response trends or structural dependencies. Deep learning models, in contrast, view the dataset holistically. While this allows them to potentially capture global patterns, it also means that their performance is more sensitive to noise, imbalance, and overfitting, particularly when true positives are scarce.

Even when the loss function was adjusted to penalize false negatives more heavily, the FNN continued to struggle with correctly identifying weak interactors. This highlights a key limitation of supervised deep learning in chemoproteomics: without sufficient signal diversity in the training data, the model fails to generalize. In contrast, IR and 4PL, retained more consistent performance across conditions, particularly when replicate data were available.

The recommendation from this benchmark analysis is to use Isotonic Regression to analyze chemoproteomics data. When used in optimal experimental design settings, this method performs just as well as four parameter logistic regression, with the addition of being unaffected by outliers due to the model's non-increasing constraint. Ultimately, these findings demonstrate the importance of method selection in chemoproteomics workflows. Deep learning models offer exciting potential but must be used with caution and preferably supplemented by models with stronger statistical guarantees under data sparsity and imbalance. Simpler curve-based methods, remain valuable for initial hit prioritization and downstream validation.

Code Availability

Access full analysis via `https://github.com/sszvetecz/ML_FinalProject.git`.

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